Task: YOLO Model for Architectural Electrical Plan Sheet Detection

Objective

We need to develop and train a YOLO-based computer vision model capable of analyzing architectural electrical plan sheets and detecting/identifying electrical symbols and components from 2D plan drawings.

The goal is to create a reliable foundation that can automatically extract electrical information from architectural/electrical PDF plan sheets and convert the visual information into structured data.

Scope of Work

1. Dataset Preparation

- Collect and organize architectural/electrical plan sheet samples.
- Convert PDF/CAD drawings into suitable image formats where required.
- Identify the electrical symbols/components that need to be detected.
- Define a clear class structure for the YOLO model.
- Annotate the required objects using a suitable annotation tool such as:
  - CVAT
  - Label Studio
  - Roboflow
- Ensure annotations are accurate and consistent.
- Split the dataset into:
  - Training

- Validation
- Testing

2. Electrical Symbol Detection

The initial model should be designed to detect common electrical plan symbols, potentially including:

- Light fixtures
- Switches
- Receptacles/outlets
- Emergency lights
- Exit signs
- Smoke detectors
- Fire alarm devices
- Electrical panels
- Junction boxes
- Data/communication outlets
- Ceiling fixtures
- Wall-mounted fixtures
- Other electrical symbols identified during dataset analysis

The exact class list should be finalized after reviewing the available plan sheets.

3. YOLO Model Training

- Select an appropriate YOLO architecture/version.
- Train the model using the annotated electrical plan dataset.
- Experiment with image resolution and augmentation strategies.
- Handle the unique challenges of architectural drawings, including:
  - Very small objects
  - Dense symbols
  - Overlapping lines
  - Different drawing scales
  - Low-resolution scans
  - Different symbol styles
  - Black-and-white drawings
  - Large plan sheets

4. Large Plan Sheet Processing

Architectural plans can be very large, so the system should support processing high-resolution plan sheets.

Investigate approaches such as:

- Image tiling
- Sliding-window detection
- Overlapping tiles
- Multi-scale detection
- Detection on high-resolution crops

The system should then combine detections from individual tiles back into the original plan-sheet coordinate system.

5. Post-Processing

Develop post-processing logic to:

- Remove duplicate detections.
- Apply confidence thresholds.
- Map detections back to original sheet coordinates.
- Count detected electrical components.
- Generate structured JSON output.

Example:


```
{
  “sheet”: “E-101”,
  “detections”: [
    {
      “type”: “light_fixture”,
      “confidence”: 0.94,
      “x”: 1250,
      “y”: 840,
      “width”: 32,
      “height”: 32
    },
```

```
  {
    “type”: “receptacle”,
    “confidence”: 0.91,
    “x”: 1420,
    “y”: 920,
    “width”: 20,
    “height”: 20
  }
 ]
}
```


6. Evaluation

Evaluate the model using standard computer-vision metrics:

- Precision
- Recall
- F1 score
- mAP@50
- mAP@50:95

Pay particular attention to small-object detection and false positives, since these are likely to be major challenges with electrical plan sheets.

Deliverables

The developer should provide:

1. Prepared/annotated dataset.
2. YOLO training configuration.
3. Training scripts/notebooks.
4. Trained model weights.
5. Validation/test results.
6. Inference script.
7. Large-PDF/plan-sheet processing pipeline.
8. Detection-to-JSON output.
9. Documentation explaining:
   - Dataset structure
   - Classes
   - Training process
   - Model performance
   - How to run inference
   - Known limitations

Preferred Technology

- Python
- Ultralytics YOLO or equivalent YOLO implementation
- OpenCV
- PyTorch
- PyMuPDF / PDF processing library
- CVAT, Label Studio, or Roboflow for annotation

Success Criteria

The initial milestone is not necessarily a production-ready model. The primary objective is to establish a working proof of concept that can take an architectural electrical plan sheet and reliably detect the defined electrical symbols.

The developer should also identify which symbols are difficult for the model and recommend what additional training data or preprocessing would be required to improve accuracy.

Important Considerations

Architectural electrical drawings are different from normal object-detection datasets. Many symbols are extremely small relative to the entire sheet, and lines, dimensions, text, walls, and other drawing elements can interfere with detection.

Therefore, tiling/high-resolution processing and dataset quality should be treated as important parts of the task, rather than simply training YOLO on full-page images.